

CREDIT RISK ANALYSIS

Power BI Dashboard & Business Insights Report

Customer Intelligence • Financial Insights • Credit Risk Analytics



Interactive Power BI portfolio project

1. Executive Summary

This project presents an interactive credit-risk dashboard built in Microsoft Power BI. It transforms loan-level information into a concise view of customer behaviour, loan portfolio characteristics, default patterns and financial risk exposure.

The dashboard contains three connected pages: a Home page for navigation, Customer & Accounts for customer/loan analysis, and Risk & Finance for risk monitoring and portfolio exposure.

Metric	Current Dashboard Value
Total Loans	39
Total Loan Amount	■472.35K
Average Interest Rate	21.5%
Default Rate	12.8%
Average Income	■71.26K
Average Loan Amount	■12.11K

2. Project Objective

The objective is to build a business-oriented lending dashboard that enables users to understand portfolio size, customer characteristics, loan distribution, repayment status, default behaviour and risk concentration.

Key analytical questions

- What is the overall size and value of the loan portfolio?
- How are loans distributed across loan grades and lending intents?
- What is the default rate of the portfolio?
- Which loan grades show higher default rates?
- How much exposure falls into each risk category?
- How does customer income relate to loan amount?
- How is loan exposure distributed across income bands?

3. Tools & Technologies

- Microsoft Power BI — dashboard development and interactive visualization.
- Power Query — data preparation and transformation.
- DAX — calculated measures and risk KPIs.
- Excel / tabular source data — underlying lending dataset.
- Data visualization & business analysis — interpretation and presentation of results.

4. Dashboard Architecture

The report follows a simple three-page navigation structure:

Page	Purpose
Home	Landing page and navigation between analytical sections.
Customer & Accounts	Customer profile, loan distribution, income and loan-status analysis.
Risk & Finance	Default rates, risk-category exposure, income-band exposure and risk KPIs.

5. Home / Navigation Page

The Home page acts as the dashboard landing screen. It introduces the Banking Customer & Risk Analytics project and provides direct navigation to Customer & Accounts and Risk & Finance.

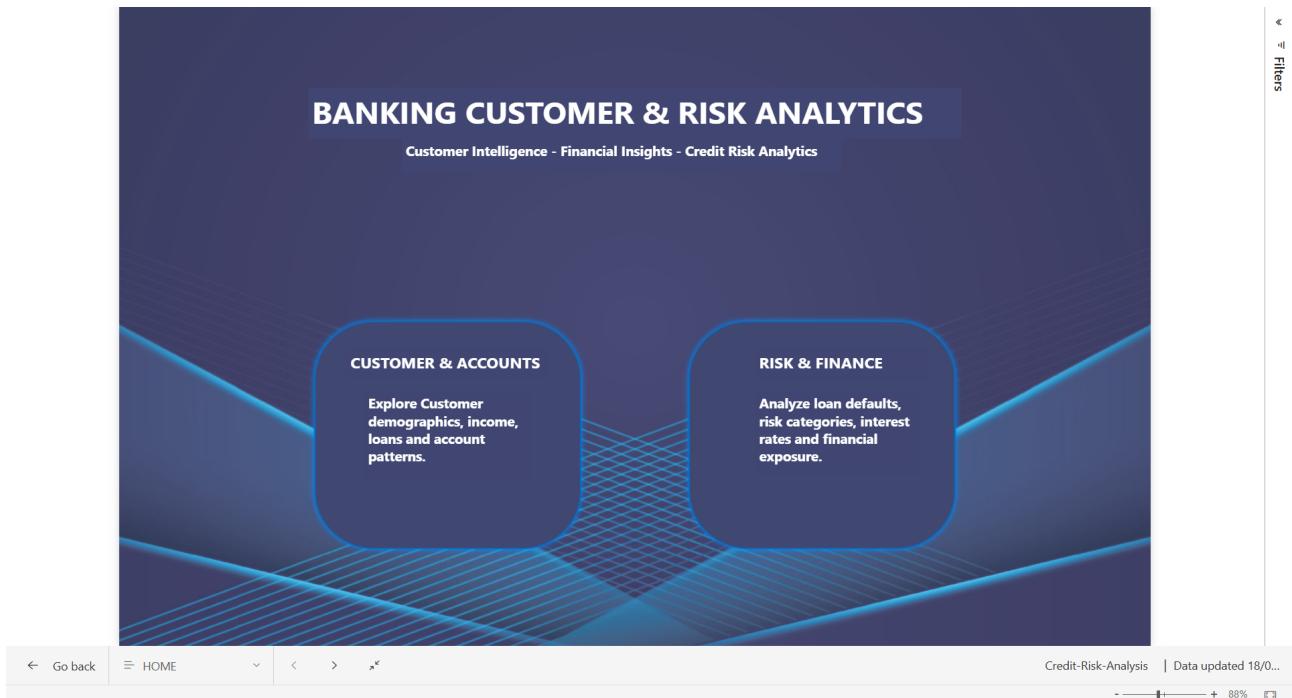


Figure 1. Clean Home page screenshot.

6. Customer & Accounts Analysis

The Customer & Accounts page provides a focused view of the selected customer/loan segment. Interactive slicers allow users to filter by Home Ownership, Loan Intent, Loan Grade and Loan Status. The page combines KPI cards with distribution, relationship and status visuals.

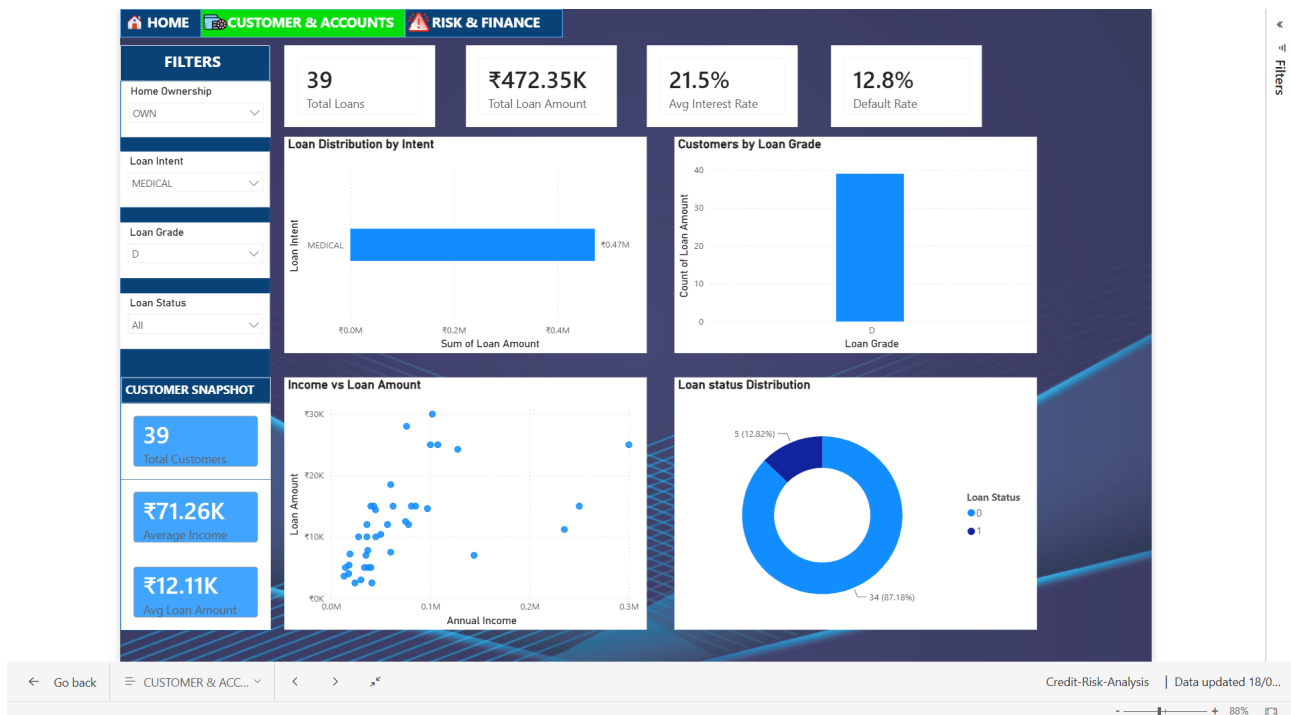


Figure 2. Clean Customer & Accounts dashboard screenshot.

Key components

- **Loan Distribution by Intent:** compares total loan amount for the selected lending intent.
- **Customers by Loan Grade:** shows customer/loan concentration by grade.
- **Income vs Loan Amount:** explores the relationship between annual income and borrowing.
- **Loan Status Distribution:** summarizes the repayment-status mix.
- **Customer Snapshot:** summarizes customers, average income and average loan amount.

7. Risk & Finance Analysis

The Risk & Finance page focuses on portfolio risk and financial exposure. It provides filters for loan grade, loan status, risk category and loan intent, together with risk KPIs and complementary visuals.

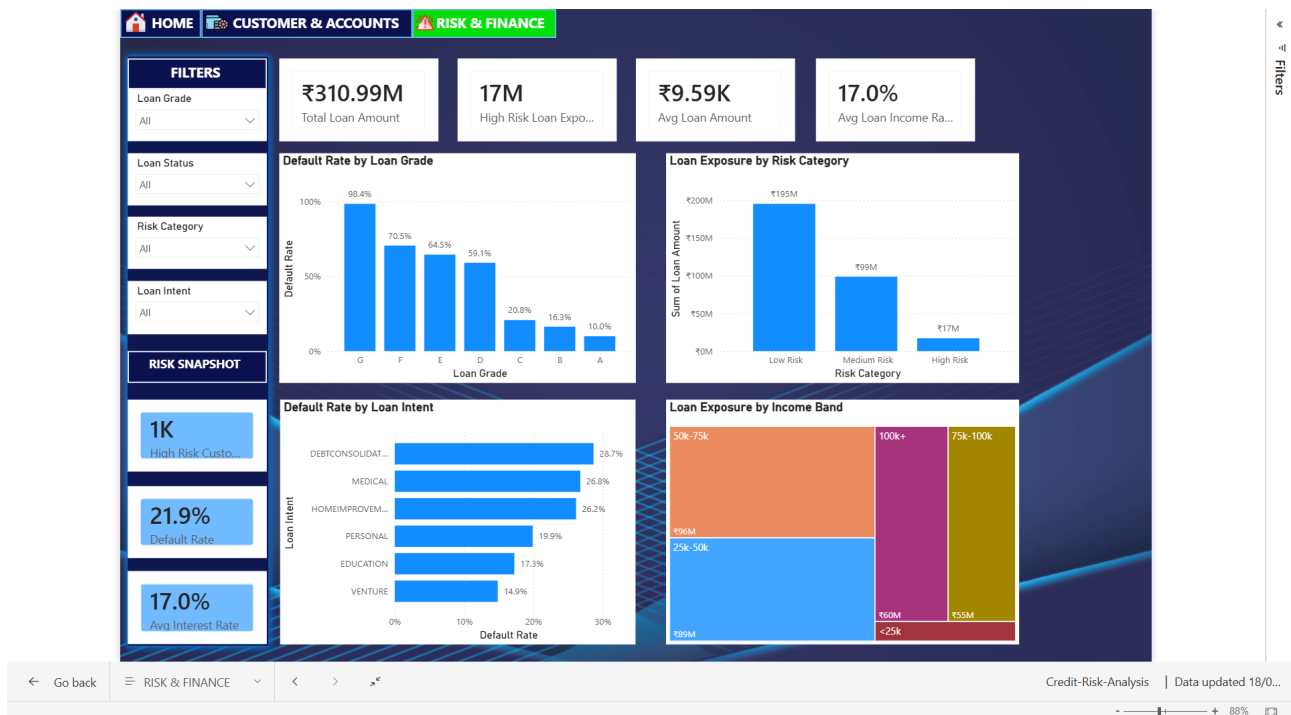


Figure 3. Clean Risk & Finance dashboard screenshot.

Key components

- **Default Rate by Loan Grade:** highlights how default rates vary across grades.
- **Loan Exposure by Risk Category:** compares total exposure across Low, Medium and High Risk.
- **Default Rate by Loan Intent:** identifies lending intents with higher default rates.
- **Loan Exposure by Income Band:** shows concentration of exposure across customer income bands.
- **Risk Snapshot:** highlights high-risk customers/exposure, default rate and average interest rate.

8. Key Findings

The current dashboard view provides the following portfolio observations. Because the Customer & Accounts page can be filtered, its displayed values should be interpreted as the selected segment rather than automatically as the full dataset.

Selected customer segment: the current Customer & Accounts view shows 39 loans and 39 total customers.

Selected-segment loan value: the current view shows total loan amount of ■472.35K and an average loan amount of ■12.11K.

Interest and default: the selected Customer & Accounts view shows an average interest rate of 21.5% and default rate of 12.8%.

Risk concentration: the Risk & Finance view shows approximately ■195M in Low Risk exposure, ■99M in Medium Risk and ■17M in High Risk in the broader dashboard context.

Grade-level risk: the Risk & Finance view shows materially higher default rates for weaker grades, with Grade G at approximately 98.4%, F at 70.5%, E at 64.5% and D at 59.1%.

Income concentration: the treemap shows meaningful loan exposure across multiple income bands, enabling the portfolio to be assessed by customer earning capacity.

9. Business Recommendations

- Prioritize monitoring of weaker loan grades because they show substantially higher default rates.
- Review high-default loan intents to understand whether underwriting criteria, pricing or borrower characteristics differ.
- Track High Risk exposure as an absolute amount as well as a percentage of the overall portfolio.
- Use income-band segmentation to identify concentrations and potential affordability risks.
- Use the dashboard's slicers during portfolio reviews to move from overall trends to specific customer segments.
- For production use, add time-based monitoring so risk and default trends can be evaluated across reporting periods.

10. Technical Implementation

The dashboard combines native Power BI visuals, calculated measures and interactive slicers. The implementation emphasizes reusable measures rather than hard-coded KPI values.

Representative DAX measures

Measure	DAX
Average Interest Rate	AVERAGE('credit_risk_dataset'[Interest Rate])
Average Loan Amount	AVERAGE('credit_risk_dataset'[Loan Amount])
High Risk Loan Exposure	CALCULATE(SUM('credit_risk_dataset'[Loan Amount]), 'credit_risk_dataset'[Risk Category] = "High Risk")

Interactivity: slicers and navigation controls allow users to move between portfolio views and examine selected segments without rebuilding the analysis.

11. Conclusion

The Credit Risk Analysis dashboard demonstrates an end-to-end Power BI analytics workflow: preparing lending data, building DAX-based KPIs, designing an interactive multi-page dashboard, and translating portfolio metrics into actionable business insights. The project is suitable for presentation as a Data Analyst portfolio project and can be extended with historical trend analysis, predictive risk scoring and automated refresh.

Recommended GitHub package: README.md + this PDF report + PBIX file + the three clean dashboard screenshots.

— End of Report —